

Mini Project R2

GROUP 1 | Biomedical Signal & Imaging Analysis

Project Focus Title: *"Predictive Diagnostics in Cardiological and Neurological Bio-Signal Analytics"*

- **Unit I EDA (Dataset 1):** Cardiovascular ECG Rhythm Variations (<https://www.kaggle.com/datasets/shayanfazeli/heartbeat>)
- **Unit I EDA (Dataset 2):** Brain Wave EEG Cognitive Load Profiles (<https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions>)
- **Unit II Regression Task:** Biomedical Lumbar Spine Geometric Measurements (<https://archive.ics.uci.edu/dataset/212/vertebral+column>)
- **Unit II Classification Task:** Wisconsin Breast Cancer Diagnostic Feature Screening (<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>)

GROUP 2 | Genomics & Oncology Diagnostics

Project Focus Title: *"High-Dimensional Biomarker Extraction and Tumor Profile Categorization Systems"*

- **Unit I EDA (Dataset 1):** Gene Expression Cancer RNA-Seq Matrices (<https://archive.ics.uci.edu/dataset/401/gene+expression+cancer+rna+seq>)
- **Unit I EDA (Dataset 2):** Hepatitis Viral Progression Markers (<https://archive.ics.uci.edu/dataset/46/hepatitis>)
- **Unit II Regression Task:** DNA Sequence Functional Variant Lengths (<https://github.com/mscbioinformatics/Biomedical-Datasets>)
- **Unit II Classification Task:** Clinical Lung Cancer Genetic Risk Screening (<https://archive.ics.uci.edu/dataset/62/lung+cancer>)

GROUP 3 | Clinical Pathology & Health Informatics

Project Focus Title: *"Clinical Parameter Optimization and Metabolic Disorder Risk Systems"*

- **Unit I EDA (Dataset 1):** Chronic Kidney Disease Laboratory Metrics (<https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease>)
- **Unit I EDA (Dataset 2):** Fetal Health Cardiotocography Matrix Profiles (<https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification>)
- **Unit II Regression Task:** Serum Biomarker Concentrations in Liver Disorders (<https://archive.ics.uci.edu/dataset/60/liver+disorders>)
- **Unit II Classification Task:** Pima Indians Diabetes Clinical Risk Mapping (<https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>)

GROUP 4 | Pharmacomedical & Life Science Logistics

Project Focus Title: *"Quantitative Structure-Activity Relationships (QSAR) and Chemical Bio-Activity Benchmarks"*

- **Unit I EDA (Dataset 1):** QSAR Aquatic Toxicity Bio-Chemical Attributes (<https://archive.ics.uci.edu/dataset/504/qsar+aquatic+toxicity>)
- **Unit I EDA (Dataset 2):** Nutritional Value Profiles of Medical Ingestibles (<https://www.kaggle.com/datasets/crawford/80-cereals>)
- **Unit II Regression Task:** Pharmacokinetic Drug Plasma Level Variations (<https://archive.ics.uci.edu/dataset/164/ozturk+gas+ischemia>)
- **Unit II Classification Task:** Wild Fungi Macro-Chemical Toxicity Separation (<https://www.kaggle.com/datasets/uciml/mushroom-classification>)

GROUP 5 | Epidemiology & Global Public Health

Project Focus Title: *"Spatiotemporal Disease Proliferation and Healthcare Delivery Resource Benchmarking"*

- **Unit I EDA (Dataset 1):** Hospitalized Patient Survival Outcome Profiles (<https://www.kaggle.com/datasets/mitishaagarwal/patient-survival-prediction-dataset>)
- **Unit I EDA (Dataset 2):** Maternal Health Risk Parameter Clusters (<https://archive.ics.uci.edu/dataset/863/maternal+health+risk+dataset>)
- **Unit II Regression Task:** Medical Insurance Premium Actuarial Pricing (<https://www.kaggle.com/datasets/mirichoi0218/insurance>)
- **Unit II Classification Task:** Heart Failure Clinical Episode Survivors Matrix (<https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records>)

GROUP 6 | E-Commerce & Digital Retail Systems

Project Focus Title: *"Consumer Engagement Architectures and Online Acquisition Models"*

- **Unit I EDA (Dataset 1):** Global Electronic Commerce Sales Transactions (<https://archive.ics.uci.edu/dataset/352/online+retail>)
- **Unit I EDA (Dataset 2):** E-Commerce Multi-Category Product Catalogs (<https://github.com/fuadhuseynov1/EDA-Exploratory-Data-Analysis>)
- **Unit II Regression Task:** E-Commerce Customer Annual Spend Tracking (<https://www.kaggle.com/datasets/iyadavvaibhav/shopping-cart-database>)
- **Unit II Classification Task:** Online Retail Customer Real-Time Purchase Intentions (<https://archive.ics.uci.edu/dataset/486/online+shoppers+purchasing+intention+dataset>)

GROUP 7 | Corporate HR Operations & Logistics Frameworks

Project Focus Title: *"Workforce Churn Mitigation Infrastructure and Delivery Performance Pipelines"*

- **Unit I EDA (Dataset 1):** Corporate Workplace Absenteeism Structural Correlates (<https://archive.ics.uci.edu/dataset/445/absenteeism+at+work>)
- **Unit I EDA (Dataset 2):** E-Commerce Supply Chain Shipping Delivery Delays (<https://www.kaggle.com/datasets/prachi13/customer-analytics>)
- **Unit II Regression Task:** Used Vehicle Marketplace Fair Value Estimations (<https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho>)
- **Unit II Classification Task:** IBM Enterprise Human Resources Employee Attrition (<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>)

GROUP 8 | Environmental Sciences & Educational Performance

Project Focus Title: *"Aerosol Proliferation Frameworks and Academic Metric Modeling Pipelines"*

- **Unit I EDA (Dataset 1):** Global Air Quality Index Matrix Metrics (<https://www.kaggle.com/datasets/hasibalmuhammad/global-air-pollution-dataset>)
- **Unit I EDA (Dataset 2):** Fast Food Menu Nutrition and Caloric Spikes (<https://www.kaggle.com/datasets/ulriktherkildsen/fast-food-nutritional-dataset>)
- **Unit II Regression Task:** Student Multi-Variable Academic Performance Indexing (<https://www.kaggle.com/datasets/mitgandhi10/dataset-for-multiple-regression>)
- **Unit II Classification Task:** Digital Network Malicious Intrusion Classification (<https://archive.ics.uci.edu/dataset/23/kdd+cup+1999+data>)

GROUP 9 | Financial Markets & Automotive Performance

Project Focus Title: *"Cryptocurrency Structural Trends and Vehicle Engine Parameter Modeling"*

- **Unit I EDA (Dataset 1):** Cryptocurrency Market Volatility & Capitalization Logs (<https://www.kaggle.com/datasets/sudalairajkumar/cryptocurrencypricehistory>)
- **Unit I EDA (Dataset 2):** Aviation Accident Logistics and Failure Diagnostics (<https://www.kaggle.com/datasets/khsamaha/aviation-accident-database-synopses>)
- **Unit II Regression Task:** Automobile Engine Configuration Fuel Efficiency (MPG) (<https://archive.ics.uci.edu/dataset/9/auto+mpg>)
- **Unit II Classification Task:** Forensic Glass Structural Fragment Material Testing (<https://www.kaggle.com/datasets/uciml/glass>)

GROUP 10 | Digital Economy & Targeted Banking Systems

Project Focus Title: *"Mobile App Ecosystem Dynamics and Direct Bank Marketing Pipelines"*

- **Unit I EDA (Dataset 1):** Google Play Store App Ratings and Engagement Patterns (<https://www.kaggle.com/datasets/lava18/google-play-store-apps>)
- **Unit I EDA (Dataset 2):** Global Video Game Industry Revenue Lifecycle Distributions (<https://www.kaggle.com/datasets/gregorut/videogamesales>)
- **Unit II Regression Task:** Industrial Power Plant Thermal Energy Output Metrics (<https://archive.ics.uci.edu/dataset/294/combined+cycle+power+plant>)
- **Unit II Classification Task:** Banking Institution Customer Term Deposit Optimization (<https://archive.ics.uci.edu/dataset/222/bank+marketing>)

GROUP 11 | Streaming Entertainment & Consumer Finance

Project Focus Title: *"On-Demand Media Streaming Demand Maps and Customer Credit Guardrails"*

- **Unit I EDA (Dataset 1):** Netflix Content Variety and International Licensing Spreads (<https://www.kaggle.com/datasets/shivamb/netflix-shows>)
- **Unit I EDA (Dataset 2):** Hospitality Infrastructure Seasonal Booking Fluctuations (<https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>)
- **Unit II Regression Task:** Urban Shared Bicycle Rental Traffic Distributions (<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>)
- **Unit II Classification Task:** Consumer Credit Card Default Risk Safeguard Matrix (<https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>)

GROUP 12 | Social Media Frameworks & Real Estate Markets

Project Focus Title: *"Global Digital Media Reach Metrics and Macro Property Valuation Grids"*

- **Unit I EDA (Dataset 1):** Global YouTube Top Creator Content Metric Trajectories (<https://www.kaggle.com/datasets/nelgiriyeewithana/global-youtube-statistics-2023>)
- **Unit I EDA (Dataset 2):** Supermarket Distribution Hub Sales Projections (<https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting>)
- **Unit II Regression Task:** Geospatial Real Estate Valuation Parameter Modeling (<https://archive.ics.uci.edu/dataset/477/real+estate+valuation+dataset>)
- **Unit II Classification Task:** Handset Performance Specifications Pricing Classifiers (<https://www.kaggle.com/datasets/iabhishekoofficial/mobile-price-classification>)

GROUP 13 | Renewable Energy & Digital Risk Management

Project Focus Title: *"Solar Radiation Harvest Modeling and E-Commerce Fraud Mitigation"*

- **Unit I EDA (Dataset 1):** Worldwide Carbon Emission Trajectories by Nation State (<https://www.kaggle.com/datasets/jmuresan/co2-emissions-by-country>)

- **Unit I EDA (Dataset 2):** Electric Vehicle Population Densities and Battery Ranges (<https://www.kaggle.com/datasets/geoffwilliams/electric-vehicle-population-data>)
- **Unit II Regression Task:** Commercial Air Flight Delay Patterns and Timing Vectors (<https://www.kaggle.com/datasets/usdot/flight-delays>)
- **Unit II Classification Task:** Digital Merchant Transaction Fraudulent Vector Screens (<https://www.kaggle.com/datasets/vbinette/ecommerce-fraud-dataset>)

GROUP 14 | Sustainable Agriculture & Telecommunication Systems

Project Focus Title: *"Agricultural Crop Production Forecasting and Telecom Churn Control Systems"*

- **Unit I EDA (Dataset 1):** Tech Industry Workplace Mental Health Survey Correlates (<https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey>)
- **Unit I EDA (Dataset 2):** Global Food Security Production Quantities Analysis (<https://www.kaggle.com/datasets/unitednations/global-food-agriculture-statistics>)
- **Unit II Regression Task:** Agricultural Harvesting Yield Performance Factors (<https://www.kaggle.com/datasets/advaypatil/crop-yield-prediction-dataset>)
- **Unit II Classification Task:** Telecom Multi-Service Subscriber Retention Pipelines (<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>)